

Aapke CRE (Chemical Reaction Engineering) ke notes mein jo graphs hain, woh **sabse zyada important hissa** hain! CRE mein bina graphs ke "Batch Reactor Data Analysis" bilkul adhoora hai. Exam mein aur PSU interviews mein in graphs se seedhe sawal aate hain. Aapko in graphs ko chhodna nahi hai, balki unhe samajhna hai kyunki ye **Order of Reaction (\$n\$)** aur **Rate Constant (\$k\$)** ki value nikalne ke sabse aasan tarike hain. Aapke notes mein mukhya roop se 4 categories ke graphs hain. Inhe aapko is tarah se padhna aur exam mein use karna hai:

1. Integral Method ke Graphs (Batch Reactor Analysis)

Integral method mein hum samay (\$t\$) aur conversion (\$X_A\$) ke beech relation nikalte hain aur use graph par plot karte hain. Agar graph mein ek seedhi line (straight line) aati hai, toh humara assume kiya gaya 'order' sahi hota hai. Exam mein equation ke sath in graphs ko zaroor draw karein:

- **Zero Order Graph:** Isme y-axis par $C_{A0}X_A$ aur x-axis par time (\$t\$) rakhte hain. Isse ek straight line milti hai jiska **Slope \$= k\$** hota hai 1.
- **First Order Graph:** Isme y-axis par $-\ln(1-X_A)$ aur x-axis par time (\$t\$) plot karte hain. Iska **Slope \$= k\$** hota hai 2, 3.
- **Second Order Graph:** Isme y-axis par $\frac{X_A}{1-X_A}$ aur x-axis par \$t\$ plot karte hain. Is seedhi line ka **Slope \$= kC_{A0}\$** hota hai 4.
- **Tip:** Aapko bas itna yaad rakhna hai ki kis order ke liye x aur y axis par kaun si values aayengi aur us line ka slope kya hoga.

2. Differential Method ka Graph

Yeh data analysis ka sabse shaktishali graph hai kyunki is ek graph se Order aur Rate Constant dono nikal aate hain 5, 6.

- **Graph Plot:** Isme y-axis par $\ln(-r_A)$ (rate of reaction ka log) aur x-axis par $\ln(C_A)$ (concentration ka log) banaya jata hai 6.
- **Kyu Important Hai?** Is graph mein banne wali straight line ka jo **Slope (ढलान)** hota hai, woh seedha **Order of reaction (\$n\$)** ke barabar hota hai. Aur is line ka jo **y-intercept (अन्तःखण्ड)** hota hai, woh $\ln k$ ki value deta hai 6.

3. Arrhenius Equation Graph (Temperature Dependency)

Jab aap Variables Affecting Rate of Reaction padhte hain, toh Temperature ke asar ko samajhne ke liye Arrhenius theory ka graph banta hai 7, 8.

- **Graph Plot:** Isme $\ln k$ ko y-axis par aur $1/T$ (inverse of temperature) ko x-axis par plot kiya jata hai 8.
- **Kyu Important Hai?** Is graph ka slope $-E/R$ hota hai (jahan E = Activation Energy aur R = Gas constant hai). Industry mein kisi bhi nayi reaction ki Activation Energy nikalne ke liye yahi graph banaya jata hai 8.

4. Size Comparison of Reactors (Levenspiel Plot)

Chapter 5 (Reactor Design) mein jab aap CSTR aur PFR ke aakar (size) ki tulna karte hain, tab ek vishesh graph banta hai 9, 10.

- **Graph Plot:** Yahan y-axis par $1/(-r_A)$ aur x-axis par Conversion (\$X_A\$) plot karte hain 10.

- **Kyu Important Hai?** Ek PFR ka volume is curve ke niche ke **Area** ke barabar hota hai, jabki CSTR ka volume ek **Rectangle** (आयत) ke area ke barabar hota hai. Is graph ko dekh kar hi pata chal jata hai ki kisi bhi positive order reaction ke liye hamesha CSTR ka volume PFR se bada kyu hota hai 10.

Aapko kya karna chahiye?

1. **Ek "Graph Sheet" banayein:** Ek single A4 paper lijiye aur us par ye sabhi mukhya graphs draw kar lijiye. Har graph ke niche uska **x-axis**, **y-axis**, aur **Slope** likh lijiye.
2. **Exam Strategy:** Jab bhi exam mein Integral ya Differential method ka sawal aaye, ya CSTR/PFR ka difference pucha jaye, toh theory ke sath-sath uska graph **zaroor** draw karein. Isse sidhe full marks milenge.

Agar aap chahein toh main in sabhi graphs aur unke equations ko ek **Slide Deck** ya **Infographic** ke roop mein compile kar sakta hoon, jisse aap inhe ek hi jagah par aasani se revise kar sakein. Kya main aapke liye yeh visual material create karoon?